

FeaGPT: figure-by-figure study and catwalk content mapping

Reference-led preparation for the catwalk manuscript | English working document | 6 October 2026

Reference and reading order

Qi, Y., Xu, R., and Chu, X. FeaGPT: an End-to-End agentic-AI for Finite Element Analysis. arXiv:2510.21993v1 (2025). <https://arxiv.org/abs/2510.21993v1>

Downloaded reference: reference/FeaGPT.pdf, 21 pages. Read the paper in its original order: Introduction; Methods A–G; Experiments and Results A–G; Conclusions. The mapping below preserves the purpose, arrangement and evidence sequence of each figure while specifying catwalk content.

Reference page previews reproduce the unchanged source pages. Source: Qi, Xu and Chu (2025), FeaGPT, arXiv:2510.21993v1, licensed CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>). The catwalk figure plans and quantitative Figure 4 are new adaptations.

Figure 1 — Workflow architecture (source page 4)

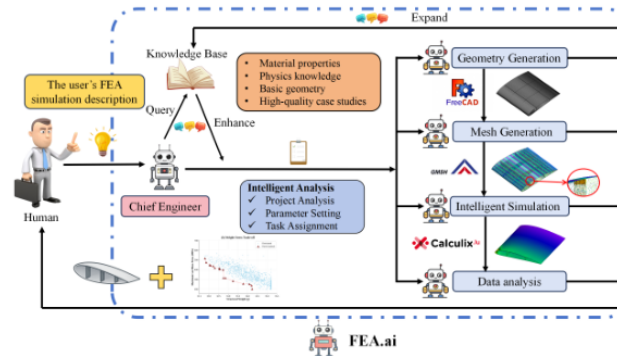


FIG. 1: FeaGPT system architecture illustrating the complete GMSA (Geometry-Mesh-Simulation-Analysis) pipeline. The framework transforms natural language specifications into validated FEA results through intelligent analysis planning, adaptive geometry generation, physics-aware meshing, and solver configuration, with optional data-driven analysis for parametric studies.

lies in semantic topology mapping: rather than extracting precise coordinates, loads and boundary conditions are specified through semantic location descriptors ("left edge", "hole boundary"), which the geometry analyzer subsequently maps to actual surface patches. Material properties are retrieved dynamically through vector search: based on contextual understanding, the system queries the knowledge base in real-time (e.g., "aerospace aluminum" automatically retrieves complete mechanical properties of Al 7075-T6). Algorithm 1 illustrates this planning process and its structured output format.

Engineering Requirements Analysis: The system analyzes engineering requirements through multi-level interpretation:

- **Quantitative specifications:** Dimensional values with unit recognition and physical consistency checking (chord length, span, thickness with mm/m/in conversion)
- **Design space definition:** Range specifications automatically expanded to complete parameter spaces for comprehensive exploration (e.g., "1.0-2.0 mm, step 0.5" generates parametric study configurations)
- **Performance objectives:** Goal statements translated to computational metrics (e.g., "lightweight" → weight minimization, "high stiffness" → displacement constraints)
- **Analysis requirements:** Problem type inference from engineering context (static, modal, fatigue, buckling)

Component Pattern Recognition: The system identifies engineering components through pattern analysis and semantic understanding, enabling intelligent geometry planning. For instance, detecting "NACA0012" versus "NACA4412" triggers appropriate airfoil profile formulations with corresponding geometric definitions.

Analysis Strategy Formulation: For comprehensive analysis planning, the system employs Gemini 2.5 Pro to formulate complete analysis strategies from natural language descriptions. The LLM comprehends engineering semantics and generates structured analysis plans containing:

- **Material specification:** Name recognition with automatic database matching and property retrieval (e.g., "Aluminum 7075-T6" → $E=71.7 \text{ GPa}$, $\nu=0.33$, $\rho=2810 \text{ kg/m}^3$, $\sigma_y=503 \text{ MPa}$)

Source page 4: Qi, Xu and Chu, FeaGPT (2025), CC BY 4.0; unchanged page preview.

What the source figure does

A user request feeds the chief engineer/planner; a knowledge base supplies geometry, materials and prior cases; four connected modules perform geometry, meshing, simulation and data analysis. Arrows show both task flow and returned results.

Catwalk panel-by-panel replacement

Keep the left-to-right request-to-result composition and the knowledge block above the workflow. Replace the aircraft request with the catwalk INP set and reference report. Use four module boxes: input/model audit; Gmsh topology reconstruction; nonlinear CalculiX execution; field recovery and report generation. Add the manifest, Skill and benchmark assets above, and the Word/PDF/CSV/VTU outputs at the return arrow.

Data to use

SKILL.md; scripts/run.py; assets/reference/manifest.json; examples/run.json

Draft catwalk caption

Figure 1. Skill-guided workflow for nonlinear static verification of the catwalk cable system, connecting versioned inputs, mesh verification, finite element solution, field recovery and engineering-report generation.

Figure 2 — Geometry and structural detail (source page 12)

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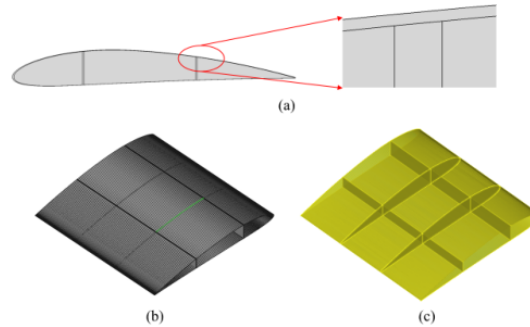


FIG. 2: Geometric details of automatically generated NACA4412 wing structure. (a) Side view showing airfoil profile with NACA4412 camber, inset detail reveals precise spar-shell interface where internal reinforcement integrates with outer skin. (b) Complete 3D assembly demonstrating spanwise ribs intersecting chordwise spars within the cambered shell. (c) Cutaway view exposing internal spar-rib framework, highlighting the automated Boolean operations that resolve complex multi-member intersections without manual intervention.

C. Mesh Generation and Quality Analysis

With all 432 geometric models successfully generated, the system proceeded to create computational meshes using Gmsh's adaptive refinement algorithms. The automated meshing module employs sophisticated adaptation strategies to handle the varying structural complexities across the parametric space. The system automatically identifies critical regions requiring refined discretization, including spar-rib intersections where stress concentrations occur, wing root connections experiencing maximum bending moments, and the leading edge requiring geometric precision.

Mesh statistics from the complete parametric study reveal exceptional consistency: element counts ranged from 119,627 to 140,578 tetrahedral elements (mean: 129,390, coefficient of variation: 4.6%), demonstrating robust scaling with structural complexity ($R^2 = 0.94$). This remarkable consistency across diverse configurations ensures reliable comparative analysis throughout the design space. The meshing process maintained deterministic behavior, producing identical meshes for repeated configurations—crucial for parametric study reproducibility.

The system successfully handled the challenging thin-walled geometry of the NACA4412 airfoil, with automated treatment of the sub-millimeter trailing edge while maintaining geometric fidelity within 0.1mm tolerance. Each configuration required approximately 34 seconds for mesh generation, enabling the complete 432-case study to be accomplished in approximately 4 hours of computational time.

Figure 3 illustrates the mesh refinement strategy employed by FeaGPT. The system automatically identified critical regions requiring higher element density—notably the spar-root junctions where maximum stresses typically occur, and the rib-spar intersections where load transfer between structural members takes place. As shown in the figure, the mesh transitions smoothly from fine elements at these critical zones to coarser elements in regions of uniform stress, optimizing computational efficiency without sacrificing accuracy. The adaptive refinement is particularly evident at the connection points between internal structures and the outer shell, where element sizes gradually transition to capture the stress gradients accurately.

While traditional shape-based quality metrics may show limitations for thin airfoil geometries due to the inherent high aspect ratios (200:1 chord-to-trailing-edge ratio), the generated meshes demonstrate excellent

Source page 12: Qi, Xu and Chu, FeaGPT (2025), CC BY 4.0; unchanged page preview.

What the source figure does

Three panels progress from an external wing view with a highlighted detail to a complete assembly and an internal structural view. The red ellipse and arrow connect global geometry to a local feature.

Catwalk panel-by-panel replacement

Retain three panels and the global-to-local zoom. Panel (a): full X-Z catwalk profile and control points. Panel (b): bottom cables, portal cables and equivalent members in three colors. Panel (c): enlarged portal/cable connection with original node and element IDs, material assignments and supports. Draw geometry directly from the INP coordinates.

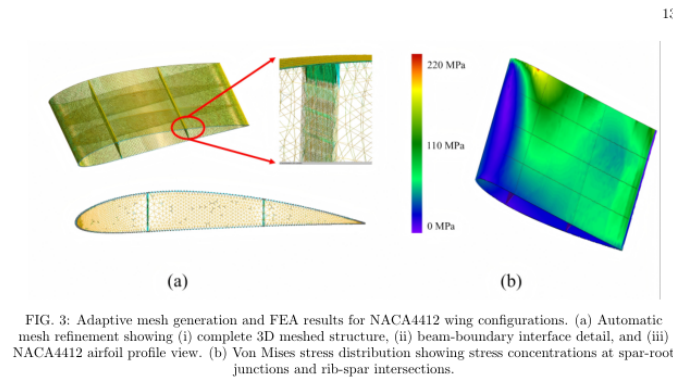
Data to use

assets/inputs/migrate_P1.inp; scripts/model.py; assets/reference/P1_input_audit.json

Draft catwalk caption

Figure 2. Catwalk finite element model: (a) longitudinal profile and control points; (b) cable families and equivalent members; (c) a portal connection with element assignments and boundary conditions.

Figure 3 — Mesh-to-field correspondence (source page 13)



validity for numerical analysis. All 432 configurations passed critical validity checks essential for reliable FEA computation:

The Jacobian determinant serves as a fundamental validity indicator for finite element meshes. Unlike shape-based metrics that penalize the high aspect ratios inherent to airfoil geometries, the Jacobian focuses on element validity for numerical computation. Our automated meshing ensures positive Jacobians throughout all configurations, guaranteeing no element inversion or folding, consistent element orientation, valid shape functions for FEA interpolation, and numerical stability in stiffness matrix assembly.

Beyond Jacobian validity, the meshes demonstrated complete topological integrity: single connected domains with no orphaned regions, complete boundary closure for proper load application, no hanging nodes or disconnected elements, and proper node-element connectivity throughout the domain. These validity metrics ensure that the automated meshing successfully produces numerically stable discretizations suitable for FEA analysis, despite the geometric challenges of thin-walled structures with extreme aspect ratios.

TABLE II: Mesh Generation Metrics and Validity Indicators

Mesh Metric	Value	Requirement
<i>Mesh Statistics</i>		
Element count range	119,627–140,578	—
Mean element count	129,390	—
Coefficient of variation	4.6%	<10%
Geometric fidelity	<0.1mm	<1mm
Generation time per case	34 seconds	—
<i>Validity Indicators</i>		
Positive Jacobian determinant	100%	100%
Minimum Jacobian	>0.001	>0
Positive element volumes	100%	100%
Topological integrity	Complete	Required
Boundary closure	Complete	Complete

Source page 13: Qi, Xu and Chu, FeaGPT (2025), CC BY 4.0; unchanged page preview.

What the source figure does

A mesh overview and local mesh enlargement occupy the left side; a stress contour occupies the right. The arrangement connects discrete geometry to the solved response.

Catwalk panel-by-panel replacement

Retain the two main panels and inset. Panel (a): original/Gmsh line mesh overlay with an enlarged tag-mapping detail. Annotate 1,125 nodes, 1,194 elements and zero coordinate discrepancy. Panel (b): P5 axial-force line contour using the same geometry; show kN, span group and governing element. Draw only line elements.

Data to use

examples/validated-run.zip: mesh/mesh_map.json and P5 field files; scripts/mesh_gmsh.py; scripts/postprocess.py

Draft catwalk caption

Figure 3. Mesh reconstruction and field transfer: (a) Gmsh mesh and original-input tag correspondence; (b) P5 bottom-cable axial-force distribution on the verified line mesh.

Figure 4 — Four-panel quantitative evidence (source page 15)

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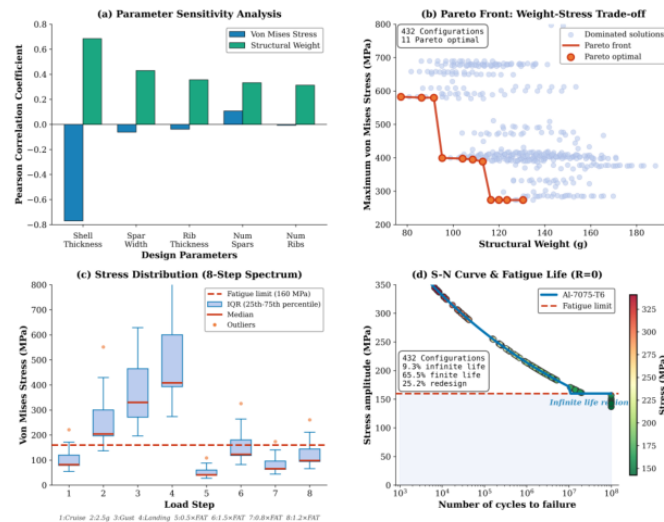


FIG. 4: Comprehensive structural analysis of 432 NACA4412 configurations. (a) Parameter sensitivity showing correlation coefficients between design variables and performance metrics. (b) Weight-stress Pareto front identifying 11 optimal configurations from the design space. (c) Stress distribution across eight-step load spectrum representing typical flight cycles (red dashed line: Al-7075-T6 fatigue limit at 160 MPa). (d) S-N curve for Al-7075-T6 with 432 configuration stress amplitudes overlaid ($R=0$ pulsating cycle).

All structural parameters showed positive correlations with weight as expected, with shell thickness having the strongest influence ($r = +0.68$, $p < 10^{-60}$), followed by spar width ($r = +0.43$) and rib thickness ($r = +0.36$).

The multi-objective optimization revealed the fundamental trade-offs inherent in aerospace structural design (Figure 4b). From the 432 tested configurations, 11 Pareto-optimal solutions were identified (2.5%), forming a clear efficiency frontier. These optimal solutions span from lightweight designs accepting higher stresses (77.3g at 582.6 MPa) to robust configurations prioritizing minimal stress at the cost of increased weight (130.5g at 273.6 MPa). The Pareto front reveals a strong weight-stress trade-off: reducing stress from 583 MPa to 274 MPa requires a 69% mass increase (77→131g), demonstrating the fundamental design conflict between minimum weight and minimum stress objectives in aerospace structures.

The eight-step load spectrum analysis (Figure 4c) reveals significant stress variation across flight conditions. The landing condition (Step 4, 3.0g) represents the design-driving load case for 88.4% of configurations (382/432), generating median stresses of 408.4 MPa with an interquartile range of 393.2–600.3 MPa. The gust condition (Step 3, 1.8g) dominates the remaining 11.6% of designs. The ground-air-ground (GAG) fatigue cycles (Steps 5–8) produce lower mean stresses (median range 40.9–122.6 MPa) but represent critical conditions for cumulative damage assessment. The wide interquartile range for high-g conditions (Steps 2–4:

Source page 15: Qi, Xu and Chu, FeaGPT (2025), CC BY 4.0; unchanged page preview.

What the source figure does

The original 2×2 panel layout moves through parameter correlations, a stress–weight Pareto scatter, load-step stress distributions and fatigue-life results. Each panel answers a different quantitative question.

Catwalk panel-by-panel replacement

Keep the 2×2 evidence layout. Panel (a): six-case CalculiX/MAPDL peak-displacement bars. Panel (b): parity scatter for 48 report/calculated span-force pairs, distinguished by cable family. Panel (c): minimum four-span safety factor by family and case with required values. Panel (d): signed relative-error heatmap for six cases × eight span groups. Use the actual summary JSON for every point.

Data to use

examples/summary.json; examples/comparison.csv; figures/figure_4_data.csv

Draft catwalk caption

Figure 4. Six-case verification: (a) maximum displacement compared with MAPDL; (b) 48 span-force comparisons with the engineering report; (c) minimum span safety factors and required values; (d) signed force differences by case and span.

Figure 5 — Grouped field visualization (source page 18)

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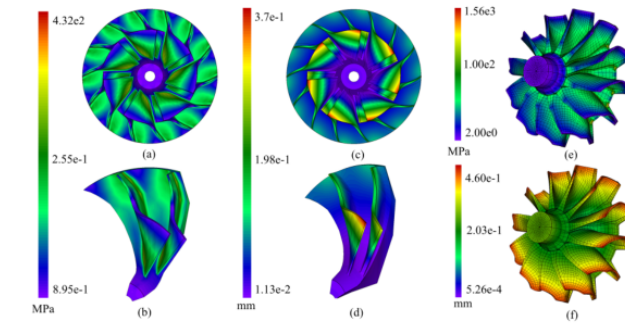


FIG. 5: FEA results from FeaGPT-generated CalculiX configurations for turbocharger components under 110,000 RPM. **Compressor (7-blade aluminum rotor)**: (a-b) von Mises stress distribution (max: 432 MPa at blade roots), (c-d) displacement field (max tip deflection: 0.37 mm). **Turbine (12-blade nickel superalloy rotor)**: (e) stress distribution (peak: 156 MPa), (f) displacement field (max: 0.46 mm). Both show proper cyclic symmetry implementation and realistic structural response.

The stress magnitudes, displacement patterns, and structural response characteristics all fall within expected ranges for turbocharger components operating under these conditions, confirming the practical engineering validity of the automated workflow.

Both CalculiX configurations were successfully generated from natural language descriptions and executed without manual intervention. The stress distributions and displacement patterns show physically realistic behavior characteristic of rotating machinery under centrifugal loading: radial blade growth with maximum deflections at blade tips, and stress concentrations at blade roots where centrifugal forces transfer to the hub. The compressor results (432 MPa maximum stress, 0.37 mm tip deflection) and turbine results (156 MPa maximum stress, 0.46 mm tip deflection) fall within expected ranges for turbocharger components operating at 110,000 RPM, validating the correctness of the automatically generated configurations. Each configuration required only 10-15 seconds of generation time, demonstrating that natural language interfaces can effectively democratize access to complex FEA tools while maintaining the technical rigor required for industrial applications.

IV. CONCLUSIONS

This work presents FeaGPT, the first natural language-driven framework that successfully automates the complete finite element analysis workflow. Through the comprehensive 432-case NACA4412 parametric study, we demonstrated the system's capability to transform natural language specifications into complete FEA results without manual intervention. The framework successfully generated and analyzed all configurations with consistent mesh quality (coefficient of variation 4.6%) and extracted meaningful stress patterns across the parameter space. Additional validation through industrial turbocharger cases—a 7-blade compressor and 12-blade turbine operating at 110,000 RPM—demonstrated successful generation of CalculiX configurations directly from natural language. Both cases executed successfully and produced physically realistic results, proving the framework's capability for industrial rotating machinery analysis.

The key contribution lies in democratizing FEA accessibility—reducing the user interaction from hundreds

Source page 18: Qi, Xu and Chu, FeaGPT (2025), CC BY 4.0; unchanged page preview.

What the source figure does

Six panels are arranged in three columns and two rows. Compressor and turbine results combine full-component/sector views and stress/displacement fields with independent color bars.

Catwalk panel-by-panel replacement

Keep six panels, arranged as three columns $\times$ two rows. Columns: P1 dead load, P5 maximum gust and P6 cooling. Upper row: signed UZ. Lower row: bottom-cable axial force. Use common limits within each field row, label mm/kN, and annotate the relevant extreme node/element. Place all six load cases in supplementary field plates using the same settings.

Data to use

examples/validated-run.zip: P1/P5/P6 nodes.csv, element results and VTU; scripts/postprocess.py

Draft catwalk caption

Figure 5. Representative field responses under dead load, maximum gust and cooling: upper row, vertical displacement; lower row, bottom-cable axial force. Columns correspond to P1, P5 and P6.

Table-by-table and algorithm mapping

Source item	Source function	Catwalk replacement
Table I, p. 11	Parameter values and configuration count	Six load combinations, step count, temperature and required K; six cases total.
Table II, p. 13	Mesh statistics and validity checks	Node/element counts, element families, coordinate discrepancy, ordered-connectivity checks and initial-stress record count.
Table III, p. 14	Solver performance and response summary	Six-case Umax, peak node, signed MAPDL difference, minimum cable factors and measured solver wall time.
Algorithm 1, p. 5	Structured JSON analysis plan	Manifest-driven task schema and ordered validate/mesh/solve/recover/compare/report stages.

Reading-to-writing sequence

I. Introduction, pp. 1–2 > I. Introduction. Paragraph 1 — Introduce agentic engineering workflows through the sequence used in FeaGPT: language-based task specification, tool execution and computational engineering results. Cite FeaGPT at the transition to finite element analysis.

II.A. System Overview, p. 3; Fig. 1 > II.A. System overview. Describe the four workflow modules and their file interfaces using Figure 1. The assistant reads the English Skill and invokes the packaged tools; each tool produces named, machine-readable evidence.

II.B. Engineering Analysis Planning and Task Orchestration, pp. 3–6; Algorithm 1 > II.B. Skill-guided planning and task orchestration. Replace the reference geometry-design JSON with the catwalk task schema: input manifest, cases P1–P6, solver identity, units, execution stages, comparison criteria and report outputs.

II.C. Intelligent Geometry Generation with Knowledge Augmentation, pp. 6–7; Fig. 2 > II.C. Versioned model reconstruction and engineering knowledge. Describe reconstruction from the original coordinates and connectivity, followed by material/section/support audit. Use Figure 2 to connect global geometry, structural families and a local connection.

II.D. Adaptive Mesh Generation, pp. 7–8; Fig. 3; Table II > II.D. Gmsh mesh generation and identity mapping. Explain point/line creation from INP IDs, section physical groups, setTransfiniteCurve(eid, 2), first-order generate(1), and MSH 4.1 export.

II.E. Automated FEA Simulation and Analysis, pp. 8–9 > II.E. Nonlinear static analysis and load-history execution. Write equilibrium as $R(u, \lambda) = f_{int}(u, \sigma_0, T) - f_{ext}(\lambda) = 0$. Describe geometric nonlinearity, initial prestress, gravity and prescribed supports within the planar X–Z system.

II.F. Intelligent Data Analysis and Batch Processing, p. 9 > II.F. Field recovery and automated verification. Define $U_{max} = \max \sqrt{U_x^2 + U_y^2 + U_z^2}$ over original nodes. Define the deformed chord n and recover $N = A_0 n^T \sigma_{max} n / 1000$ in kN, averaging the eight integration-point tensors.

II.G. Scalable Batch Processing Architecture, pp. 9–10 > II.G. Reproducible batch execution and report generation. Describe independent case directories, common geometry audit, recorded solver commands, failure preservation and per-stage status.

III.A. Natural Language Input and System Understanding, p. 11; Table I > III.A. Task specification and six-case execution. Present the task request and its mapped case/output specification. Table I enumerates the six combinations and required safety factors.

III.B. Geometry Generation Results, pp. 11–12; Fig. 2 > III.B. Model reconstruction results. Report 1,125 original nodes, 1,123 T3D2 cable elements and 71 B31 equivalent members. Show the complete geometry and

member assignments in Figure 2.

III.C. Mesh Generation and Quality Analysis, pp. 12–13; Table II; Fig. 3 > III.C. Mesh identity and mapping results. Report identical node/element counts, zero coordinate discrepancy and matching ordered connectivity between the input and the Gmsh mesh. Use the local inset to display the mapping.

III.D. FEA Simulation and Results Analysis, p. 14; Table III > III.D. Nonlinear response and benchmark comparison. Report completion of all six cases and matching peak-displacement node IDs. Present the six numerical rows in Table III.

III.E. Intelligent Result Analysis and Autonomous Design Evaluation, pp. 14–16; Fig. 4 > III.E. Cable response interpretation and safety assessment. Follow the four panels in order: global displacement agreement, span-force agreement, cable safety factors and the spatial/case distribution of signed differences.

III.F. Computational Performance and Scalability, pp. 16–17 > III.F. Execution performance and automation accounting. Report the recorded solver wall times by case and their sum, with Python/platform/solver identity. Label timing boundaries explicitly.

III.G. Industrial Validation: CalculiX Configuration Generation, pp. 17–18; Fig. 5 > III.G. Engineering case presentation and report delivery. Present Figure 5 as the engineering field plate: three representative cases and two response quantities. Compare shared field scales within each row.

IV. Conclusions, pp. 18–19 > IV. Conclusions. Conclusion 1 — Summarize the Skill-guided sequence connecting versioned bridge inputs to mesh verification, nonlinear solution and engineering-report generation.